

Network Subnetting Plan & Guide

Comprehensive breakdown of 192.168.10.0/24 divided into 4 subnets with device assignments and step-by-step logic.

1. Subnet Breakdown (192.168.10.0 /24 into 4 Subnets)

To divide the original /24 network into 4 equal subnets, we borrow 2 host bits ($2^2 = 4$). This changes the subnet prefix from /24 to /26, yielding a subnet mask of 255.255.255.192 with 64 addresses per subnet (62 usable hosts).

Subnet	Network Address	Subnet Mask	Usable IP Range	Broadcast Address
Subnet 1	192.168.10.0	255.255.255.192	192.168.10.1 – 192.168.10.62	192.168.10.63
Subnet 2	192.168.10.64	255.255.255.192	192.168.10.65 – 192.168.10.126	192.168.10.127
Subnet 3	192.168.10.128	255.255.255.192	192.168.10.129 – 192.168.10.190	192.168.10.191
Subnet 4	192.168.10.192	255.255.255.192	192.168.10.193 – 192.168.10.254	192.168.10.255

2. Device IP Address Assignments (Subnet 1)

For demonstration, we select Subnet 1 (192.168.10.0/26, usable range 192.168.10.1 – 192.168.10.62) to connect 5 devices. Network and broadcast addresses are strictly reserved.

Device Type	Assigned IP Address	Subnet Mask	Default Gateway
Router (Gateway)	192.168.10.1	255.255.255.192	N/A
Laptop 1	192.168.10.10	255.255.255.192	192.168.10.1
Laptop 2	192.168.10.11	255.255.255.192	192.168.10.1
Network Printer	192.168.10.20	255.255.255.192	192.168.10.1
Mobile Phone	192.168.10.30	255.255.255.192	192.168.10.1

3. Subnet Plan & Network Diagram

Visual topology showing Subnet 1 architecture and device connectivity:

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+-----+
| Subnet 1: 192.168.10.0/26 (Mask: 255.255.255.192) |
| Range: 192.168.10.1 to 192.168.10.62 (Broadcast: 192.168.10.63) |
+-----+
|
+--- [ Router / Gateway ] -----> IP: 192.168.10.1
|
+--- [ Switch / Access Point ]
|
+-- Laptop 1 .....> IP: 192.168.10.10
+-- Laptop 2 .....> IP: 192.168.10.11
+-- Network Printer .....> IP: 192.168.10.20
+-- Mobile Device .....> IP: 192.168.10.30

```

4. Step-by-Step Subnetting Explanation (Beginner's Guide)

Hey friend! Think of an IP address like a mailing address. It has a *neighborhood* (network part) and a *house number* (host part). Subnetting is simply taking a big neighborhood and splitting it up into 4 smaller, organized gated communities!

Step 1: Finding the Subnet Mask

We started with **192.168.10.0/24**. The **/24** means the first 24 bits belong to the network (neighborhood). Since we need **4 subnets**, we ask: *How many bits do we need to flip to get 4 combinations?* Using binary math, $2^2 = 4$, so we borrow **2 bits** from the host side.

Adding 2 bits to 24 gives us **/26**. In binary, that extra 2 bits turned on in the last octet gives us **11000000**, which equals **192**. Thus, our new subnet mask is **255.255.255.192**.

Step 2: Calculating Hosts & Block Sizes

With 6 bits left for hosts ($32 - 26 = 6$), each subnet can hold $2^6 = 64$ **total addresses**. Because the first address is the network ID and the last is the broadcast address, we subtract 2, leaving us with **62 usable host addresses** per subnet.

Our block size jumps by 64: **0, 64, 128, and 192**. That's how we get our 4 distinct subnet ranges!

Step 3: Assigning Device IP Addresses

When giving IPs to our 5 devices in Subnet 1 (**192.168.10.0** to **192.168.10.63**), we followed two golden rules:

- **Never use the Network ID** (**192.168.10.0**) — this is the name of the subnet itself.

- **Never use the Broadcast Address** (*192.168.10.63*) — this shouts messages to every device at once.

We assigned the router to *192.168.10.1* (the standard gateway convention), and the remaining devices to clean, sequential addresses (*.10*, *.11*, *.20*, *.30*) within the usable range.